

The Zeta–Weil Form and Nonunital GNS Transfer

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Abstract

The full zeta–Weil functional is written on a concrete nonunital convolution $*$ -algebra, with its polar, archimedean, and prime-power terms fixed. Weil’s theorem identifies positivity on convolution squares with the Riemann hypothesis. We prove the algebraic GNS theorem in the nonunital setting, including automatic closability and the exact bounded multiplier criterion, and show that dagger transfer extends isometrically to Hilbert completions. Positive unitization and approximate identities are treated separately under C^* -hypotheses. Consequently, a positive dagger realization of the Weil functional would prove the Riemann hypothesis only when its positivity is obtained independently of the zeros or of Weil positivity itself. Constructing such a realization remains open.

1 The exact zeta–Weil form

The decisive question is whether the arithmetic expression below is a vector norm for a reason that does not already assume the location of the zeros. We first fix the expression completely.

All integrals on $\mathbb{R}$ use Lebesgue measure. Set

$$\mathcal{A}_\zeta = \left\{ f \in C^\infty(\mathbb{R}) : \begin{array}{l} \text{there is an } \epsilon > 0 \text{ such that} \\ e^{(1/2+\epsilon)|u|} f^{(k)}(u) \in L^1(\mathbb{R}, du) \quad \text{for every } k \geq 0 \end{array} \right\}.$$

For $f, g \in \mathcal{A}_\zeta$, define

$$(f \star g)(u) = \int_{\mathbb{R}} f(v)g(u-v) dv, \quad f^*(u) = \overline{f(-u)}, \quad (1.1)$$

and use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(u)e^{izu} du, \quad f(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(t)e^{-itu} dt. \quad (1.2)$$

The inverse formula in (1.2) is asserted for real u and integrates with respect to dt .

Proposition 1.1. *The space $\mathcal{A}_\zeta$ is a commutative nonunital $*$ -algebra under (1.1). If f has exponent ϵ , then $\widehat{f}$ is holomorphic on $|\operatorname{Im} z| < 1/2 + \epsilon$. In every common strip of holomorphy,*

$$\widehat{f \star g}(z) = \widehat{f}(z)\widehat{g}(z), \quad \widehat{f^*}(z) = \overline{\widehat{f}(\bar{z})}. \quad (1.3)$$

In particular,

$$\widehat{f^* \star f}(t) = |\widehat{f}(t)|^2 \quad (t \in \mathbb{R}). \quad (1.4)$$

Proof. Choose $0 < \delta < \min(\epsilon_f, \epsilon_g)$ and put $a = 1/2 + \delta$. The inequality $e^{a|u|} \leq e^{a|v|}e^{a|u-v|}$ and the weighted L^1 convolution inequality show that

$$e^{a|\cdot|}(f \star g)^{(k)} = e^{a|\cdot|}(f^{(k)} \star g) \in L^1(\mathbb{R})$$

for every k . The same estimates justify Fubini's theorem in (1.3); the assertion for the involution follows by $u \mapsto -u$. Weighted integrability of all derivatives gives Fourier inversion and rapid decay of $\widehat{f}$ on each closed substrip.

If $e \in \mathcal{A}_\zeta$ were a convolution identity, convolution with a Gaussian and (1.3) would give $\widehat{e}(t) = 1$ for every real t . This contradicts the Riemann–Lebesgue lemma, since $e \in L^1(\mathbb{R})$. $\square$

Let

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2), \quad \psi(s) = \frac{\Gamma'(s)}{\Gamma(s)}.$$

Thus the archimedean normalization is fixed by

$$2 \operatorname{Re} \frac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}} \left(\frac{1}{2} + it \right) = \operatorname{Re} \psi \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi. \quad (1.5)$$

Write $\Lambda_{\text{vM}}(n) = \log p$ when $n = p^m$ with p prime and $m \geq 1$, and $\Lambda_{\text{vM}}(n) = 0$ otherwise. Define the linear functional $L_\zeta : \mathcal{A}_\zeta \rightarrow \mathbb{C}$ by

$$\begin{aligned} L_\zeta(g) &= \widehat{g}(i/2) + \widehat{g}(-i/2) && \text{(polar)} \\ &+ \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(t) \left[\operatorname{Re} \psi \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi \right] dt && \text{(archimedean)} \\ &- \sum_{n \geq 2} \frac{\Lambda_{\text{vM}}(n)}{\sqrt{n}} (g(\log n) + g(-\log n)) && \text{(prime powers)}. \end{aligned} \quad (1.6)$$

The centered Mellin parameter in these conventions is

$$\widehat{g}(-i(s - \tfrac{1}{2})) = \int_0^\infty g(\log x) x^{s-1/2} \frac{dx}{x}. \quad (1.7)$$

Equation (1.7) is the logarithmic form of Weil's Mellin convention [1, p. 256]; functions in $\mathcal{A}_\zeta$ satisfy his conditions (A) and (B) on p. 259.

Proposition 1.2. *Every term in (1.6) converges, and*

$$L_\zeta(g^*) = \overline{L_\zeta(g)}. \quad (1.8)$$

If ρ runs over the nontrivial zeros of $\zeta(s)$, counted with multiplicity, then

$$L_\zeta(g) = \sum_{\rho} \widehat{g}(-i(\rho - \tfrac{1}{2})), \quad (1.9)$$

and the sum is absolutely convergent.

Proof. For a suitable $\epsilon > 0$, weighted $W^{1,1}$ estimates on each half-line give

$$|g(u)| \leq C e^{-(1/2+\epsilon)|u|}.$$

The prime-power series is consequently dominated by $C \sum_{n \geq 2} \Lambda_{\text{vM}}(n) n^{-1-\epsilon}$. Repeated integration by parts makes $\widehat{g}(t)$ rapidly decreasing on $\mathbb{R}$, whereas the bracket in (1.6) is $O(\log(2 + |t|))$. This proves convergence of the local terms.

For real t , the bracket is real and even, while $\widehat{g}^*(t) = \overline{\widehat{g}(t)}$; the two polar values and the two prime-power values are interchanged by the involution. This proves (1.8).

Formula (1.9) is the $k = \mathbb{Q}$, trivial-character specialization of Weil's formula (11), not a formula printed there verbatim in the normalization (1.2)–(1.7); formula (11) and the ensuing positivity lemma are on p. 262 of [1]. Repeated integration by parts is uniform for $|\text{Im } z| \leq 1/2$, and the standard zero count is $O(T \log T)$. Hence the zero sum is absolute for the present test class. $\square$

The calculation that controls the question is now immediate.

Theorem 1.3 (Weil's positivity criterion). *With the conventions above,*

$$\text{RH} \iff L_\zeta(f^* \star f) \geq 0 \quad \text{for every } f \in \mathcal{A}_\zeta. \quad (1.10)$$

Under RH, writing each zero as $\rho = 1/2 + i\gamma_\rho$, one has the exact norm calculation

$$L_\zeta(f^* \star f) = \sum_{\rho} |\widehat{f}(\gamma_\rho)|^2. \quad (1.11)$$

Proof. If RH holds, substitute $\rho = 1/2 + i\gamma_\rho$ in (1.9) and use (1.4); this gives (1.11). Conversely, an off-line zero is separated by Weil's polynomial–Gaussian construction [1, p. 263], which makes the zero side of (1.9) negative on a convolution square. The inverse Fourier transforms used there are polynomial–Gaussian functions and therefore belong to $\mathcal{A}_\zeta$. Thus positivity on the class specified here implies RH. $\square$

The right side of (1.11) explains positivity conditional on RH; it cannot serve as an independent construction of the norm, because both the real parameters γ_ρ and their existence in that form use the asserted zero placement.

2 Algebraic GNS without a unit

Let $\mathfrak{A}$ be a complex $*$ -algebra, with no norm or unit assumed, and let $W : \mathfrak{A} \rightarrow \mathbb{C}$ be positive:

$$W(a^*a) \geq 0 \quad (a \in \mathfrak{A}).$$

Theorem 2.1 (Nonunital algebraic GNS). *Put*

$$\langle a, b \rangle_W = W(b^*a), \quad \mathcal{N}_W = \{a : W(a^*a) = 0\}.$$

Then $\mathcal{N}_W$ is a left ideal and

$$D_W = \mathfrak{A}/\mathcal{N}_W, \quad \pi_W(a)[b] = [ab], \quad (2.1)$$

is a $$ -representation on a pre-Hilbert space. Regard D_W as a common invariant dense domain in its Hilbert completion $\mathcal{H}_W$. For every $a \in \mathfrak{A}$,*

$$\pi_W(a^*) \subseteq \pi_W(a)^*, \quad (2.2)$$

so $\pi_W(a)$ is closable. It extends boundedly to $\mathcal{H}_W$ if and only if there is a constant C_a such that

$$W(b^*a^*ab) \leq C_a^2 W(b^*b) \quad (b \in \mathfrak{A}). \quad (2.3)$$

The least such C_a is the norm of the extension.

Proof. Positivity and polarization make $\langle \cdot, \cdot \rangle_W$ Hermitian and give Cauchy–Schwarz. If $b \in \mathcal{N}_W$, then

$$W((ab)^*ab) = \langle a^*ab, b \rangle_W = 0,$$

so $\mathcal{N}_W$ is a left ideal and (2.1) is well defined. For classes in D_W ,

$$\langle \pi_W(a)[b], [c] \rangle_W = W(c^*ab) = \langle [b], \pi_W(a^*)[c] \rangle_W.$$

This proves (2.2). Its right side has dense domain D_W , hence the adjoint of $\pi_W(a)$ is densely defined and $\pi_W(a)$ is closable. Finally,

$$\|\pi_W(a)[b]\|^2 = W(b^*a^*ab),$$

so the bounded-operator criterion is exactly (2.3). $\square$

For $a = a^*$, the operator $\pi_W(a)$ is symmetric and closable. Neither essential self-adjointness nor even the existence of a self-adjoint extension follows. For example,

$$P = -i \frac{d}{dx}, \quad \mathcal{D}(P) = C_c^\infty(0, \infty) \subset L^2(0, \infty),$$

is symmetric and closable, but its deficiency indices are $(1, 0)$, so it has no self-adjoint extension. More generally, closing the individual operators in (2.1) does not identify a common domain on which their products remain the given algebra products. The minimal closed representation uses the common graph domain, as in [3, Definition 2.4 and Lemma 2.6].

Example 2.2 (An unbounded positive GNS multiplier). Let $\mathfrak{A} = \mathbb{C}[x]$, let $x^* = x$, and set

$$W(p) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} p(t) e^{-t^2} dt.$$

Then W is positive, but multiplication by x is unbounded, since

$$\frac{\|\pi_W(x)[x^n]\|^2}{\|[x^n]\|^2} = \frac{\int_{\mathbb{R}} t^{2n+2} e^{-t^2} dt}{\int_{\mathbb{R}} t^{2n} e^{-t^2} dt} = n + \frac{1}{2}.$$

3 Dagger transfer and completion

Theorem 3.1 (Dagger transfer). *Let $T : \mathfrak{A} \rightarrow \mathfrak{B}$ be a $*$ -homomorphism of complex $*$ -algebras, let $\omega : \mathfrak{B} \rightarrow \mathbb{C}$ be positive, and put $W = \omega \circ T$. Then*

$$\langle a, c \rangle_W = \langle T(a), T(c) \rangle_\omega, \quad \mathcal{N}_W = T^{-1}(\mathcal{N}_\omega). \quad (3.1)$$

Consequently

$$U_0 : D_W \longrightarrow D_\omega, \quad [a] \longmapsto [T(a)], \quad (3.2)$$

is an isometric injection and extends uniquely to an isometry

$$U : \mathcal{H}_W \longrightarrow \mathcal{H}_\omega \quad (3.3)$$

with closed range. On the algebraic common domain, and only there without further hypotheses,

$$U_0 \pi_W(a) = \pi_\omega(T(a)) U_0. \quad (3.4)$$

If $\mathfrak{B}$ is a C^* -algebra, both sides of (3.4) extend boundedly and

$$\|\pi_\omega(b)\| \leq \|b\|, \quad \|\pi_W(a)\| \leq \|T(a)\|, \quad (3.5)$$

while

$$U \pi_W(a) = \pi_\omega(T(a)) U \quad \text{on } \mathcal{H}_W. \quad (3.6)$$

Proof. For $a, c \in \mathfrak{A}$,

$$W(c^*a) = \omega(T(c)^*T(a)),$$

which proves positivity, (3.1), and the isometry (3.2). Every isometry of pre-Hilbert spaces extends uniquely to their completions; its range is closed because its domain is complete. Multiplicativity of T gives (3.4).

Suppose now that $\mathfrak{B}$ is a C^* -algebra. The C^* -order inequality gives

$$\omega(d^*b^*bd) \leq \|b\|^2 \omega(d^*d),$$

which proves the first estimate in (3.5). Taking $b = T(a)$ and $d = T(c)$ proves the second. Continuity then extends (3.4) to (3.6). No equality between arbitrary unbounded closure domains is used. $\square$

4 Positive unitization for nonunital C^* -algebras

The unit missing from a convolution algebra cannot be adjoined to an arbitrary positive functional without an additional bound. For a nonunital $*$ -algebra $\mathfrak{A}$, write $\mathfrak{A}^+ = \mathfrak{A} \oplus \mathbb{C}1$ and

$$\omega_m(a + \lambda 1) = \omega(a) + m\lambda. \quad (4.1)$$

Direct minimization in λ gives the algebraic criterion

$$\omega_m \text{ is positive} \iff \begin{cases} m \in \mathbb{R}, & m \geq 0, \\ \omega(a^*) = \overline{\omega(a)}, \\ |\omega(a)|^2 \leq m \omega(a^*a) & (a \in \mathfrak{A}). \end{cases} \quad (4.2)$$

Theorem 4.1 (The nonunital C^* case). *Let $\mathfrak{A}$ be a nonunital C^* -algebra and let $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ be positive. Then ω is Hermitian and bounded, and the functional (4.1) on the standard unitization is positive exactly when*

$$m \in \mathbb{R}, \quad m \geq \|\omega\|. \quad (4.3)$$

Let (e_i) be any positive contractive two-sided approximate identity. For the minimal extension $m = \|\omega\|$, its GNS vector $\Omega = [1]$ satisfies

$$[e_i] \longrightarrow \Omega, \quad (4.4)$$

and the restricted representation of $\mathfrak{A}$ is bounded, nondegenerate, and cyclic:

$$\omega(a) = \langle \pi_\omega(a)\Omega, \Omega \rangle, \quad \overline{\pi_\omega(\mathfrak{A})\Omega} = \mathcal{H}_\omega. \quad (4.5)$$

For $m > \|\omega\|$, the unit vector of that larger extension is not the limit in (4.4).

Proof. Every self-adjoint element is the difference of two positive elements, so positivity makes ω Hermitian. If ω were unbounded, its values would be unbounded on the positive unit ball: otherwise the decomposition of an arbitrary element into four positive parts would bound it. One could then choose $a_n \geq 0$ with

$$\|a_n\| \leq 2^{-n}, \quad \omega(a_n) = 1.$$

The norm-convergent positive sum $a = \sum_n a_n$ would satisfy $\omega(a) \geq \sum_{n=1}^N \omega(a_n) = N$ for every N , a contradiction. Hence ω is bounded.

Positive contractive approximate identities exist. Cauchy–Schwarz and $e_i a \rightarrow a$ give

$$|\omega(a)|^2 = \lim_i |\omega(e_i a)|^2 \leq \|\omega\| \omega(a^* a). \quad (4.6)$$

Equations (4.2) and (4.6) prove positivity for $m = \|\omega\|$; adding $(m - \|\omega\|)$ times the positive quotient character $a + \lambda 1 \mapsto \lambda$ proves sufficiency for every larger m . Conversely, a positive functional on a unital C^* -algebra has norm equal to its value at the unit, so restriction of ω_m gives $\|\omega\| \leq m$.

It remains to identify the unit vector. If $\|\omega\| > 0$, choose $\|a\| \leq 1$ with $|\omega(a)|$ arbitrarily close to $\|\omega\|$. Using $e_i a \rightarrow a$ and Cauchy–Schwarz shows

$$\omega(e_i^2) \rightarrow \|\omega\|.$$

Since $0 \leq e_i^2 \leq e_i \leq 1$ in the unitization, also $\omega(e_i) \rightarrow \|\omega\|$. Therefore

$$\|[1] - [e_i]\|^2 = \|\omega\| - 2\omega(e_i) + \omega(e_i^2) \rightarrow 0,$$

which is (4.4); the zero functional is immediate. Moreover $\pi_\omega(a)\Omega = [a]$, proving cyclicity, and $\pi_\omega(e_i)[a] = [e_i a] \rightarrow [a]$, proving nondegeneracy. The C^* -estimate in (3.5) proves boundedness. If $m > \|\omega\|$, the same computation instead gives

$$\lim_i \|[1] - [e_i]\|^2 = m - \|\omega\| > 0.$$

□

Segal’s original formulation defines states in the possibly nonunital case on p. 75, states Theorem 1 and constructs an approximate identity on pp. 76–77, proves the quotient and multiplier bound on p. 78, and obtains the cyclic vector on pp. 78–79 of [2].

Example 4.2 (Positive unitization can fail algebraically). Let $\mathfrak{A} = c_{00}(\mathbb{N})$ with pointwise multiplication and conjugation, and define

$$\omega(a) = \sum_{n \geq 1} n a_n.$$

This is a Hermitian positive functional. If a positive extension ω_m existed, Cauchy–Schwarz in the unitization, applied to the n th point mass e_n , would give

$$n^2 = |\omega_m(e_n)|^2 \leq \omega_m(1) \omega_m(e_n^* e_n) = mn$$

for every n , which is impossible for finite m .

5 Why an indefinite quotient does not give positivity

Let $V = \mathbb{C}^2$, $J = \text{diag}(1, -1)$, and

$$[x, y] = y^* J x.$$

The vectors $e_1 + e_2$ and $e_1 - e_2$ are isotropic, while their sum is not. Thus the isotropic cone $\{x : [x, x] = 0\}$ is not a linear subspace and cannot be divided out. The radical $\{x : [x, y] = 0 \ \forall y\}$ is zero here. In general, quotienting by the radical produces only a nondegenerate Hermitian space; it supplies neither a fundamental symmetry nor a complete Hilbert majorant. In this finite-dimensional example such data can be chosen, but they are extra data, not consequences of the radical quotient.

Nor is a vector functional for an indefinite adjoint positive. On $M_2(\mathbb{C})$ put $A^\sharp = J A^* J$, let $\Omega = e_1$, and set $\omega(A) = [A\Omega, \Omega]$. For

$$X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

one has

$$\omega(X^\sharp X) = [X\Omega, X\Omega] = [e_2, e_2] = -1. \quad (5.1)$$

Hence an indefinite pairing does not provide the positivity required in (1.10).

6 The independent arithmetic question

Proposition 6.1 (Formal factorization). *For a linear functional W on a complex $*$ -algebra $\mathfrak{A}$, the following are equivalent:*

1. W is positive;
2. there are a complex $*$ -algebra $\mathfrak{B}$ and maps

$$\omega : \mathfrak{B} \longrightarrow \mathbb{C}, \quad T : \mathfrak{A} \longrightarrow \mathfrak{B},$$

where ω is positive, T is a $*$ -homomorphism, and $W = \omega \circ T$.

Proof. Dagger transfer proves (2) $\Rightarrow$ (1). For the converse take $\mathfrak{B} = \mathfrak{A}$, $\omega = W$, and T equal to the identity. $\square$

For $W = L_\zeta$, formal factorization is therefore equivalent to the positivity in (1.10), and hence to RH. The identity factorization, or a Hilbert space whose spectral data are prescribed from the zeta zeros, cannot establish that positivity independently.

Open Problem 6.2 (Independent dagger realization of L_ζ). Construct a complex $*$ -algebra $\mathfrak{B}$, a positive linear functional $\omega : \mathfrak{B} \rightarrow \mathbb{C}$, and a $*$ -homomorphism $T : \mathcal{A}_\zeta \rightarrow \mathfrak{B}$ such that

$$L_\zeta = \omega \circ T. \quad (6.1)$$

Derive the carrier, the map, and the positivity of ω from the polar, archimedean, and prime-power structures in (1.6), without using RH, the multiset of zeta zeros, or an assertion equivalent to (1.10). Prove (6.1) with all three local terms and the Fourier and measure conventions above. A solution with $\mathfrak{B}$ a C^* -algebra would additionally give the bounded realization (3.5)–(3.6).

References

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